

(TR-104) PYTHON WITH AI/ML

Training Day 31 Report

Date: 23 February 2026

Topic Covered: Model Optimization, Production Readiness & Introduction to Transformers

The thirty-first day of training focused on improving the performance and production readiness of NLP systems along with understanding the limitations of classical word embeddings and the fundamentals of transformer-based models. The training also included practical implementation using the Hugging Face Transformers library.

- Evaluated NLP model performance by analyzing accuracy, precision, recall, response quality, and response time.
- Studied retrieval quality and similarity score evaluation for embedding-based systems.
- Explored optimization techniques such as preprocessing improvements, similarity threshold tuning, and hyperparameter tuning.
- Learned the importance of caching in production systems and understood Redis-based caching concepts for reducing response time.
- Implemented logging concepts for FastAPI applications including request logging, error logging, and monitoring.
- Studied API validation and error handling using Pydantic models and exception handling techniques.
- Understood security concepts including JWT authentication, CORS configuration, and rate limiting for APIs.
- Learned deployment preparation concepts including Dockerfile creation, requirements.txt generation, and containerized application setup.
- Understood the limitations of TF-IDF, Word2Vec, GloVe, and FastText embeddings.
- Learned why transformers were introduced for contextual understanding in NLP systems.
- Studied the concept of contextual embeddings and how the same word can have different meanings depending on context.

- Learned the fundamentals of self-attention, Query-Key-Value mechanism, and multi-head attention.
- Understood the difference between encoder-only, decoder-only, and encoder-decoder transformer architectures.
- Explored the Hugging Face ecosystem including tokenizer, model, pipeline, trainer, and model hub.
- Implemented sentiment analysis using Hugging Face transformers pipeline.
- Performed manual inference using AutoTokenizer and AutoModelForSequenceClassification.
- Analyzed input IDs, logits, hidden states, and contextual embeddings generated by transformer models.
- Compared transformer-based contextual embeddings with static Word2Vec embeddings.
- Observed how transformers generate different embeddings for the same word based on sentence context.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/resume_classifier
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/from-ml-model-to-rest-api-serving-a-resume-classifier-with-fastapi-734a6fd886f6>

Overall, the training helped in strengthening understanding of production-ready NLP systems, API optimization, deployment preparation, transformer architectures, contextual embeddings, and modern NLP workflows using Hugging Face transformers.